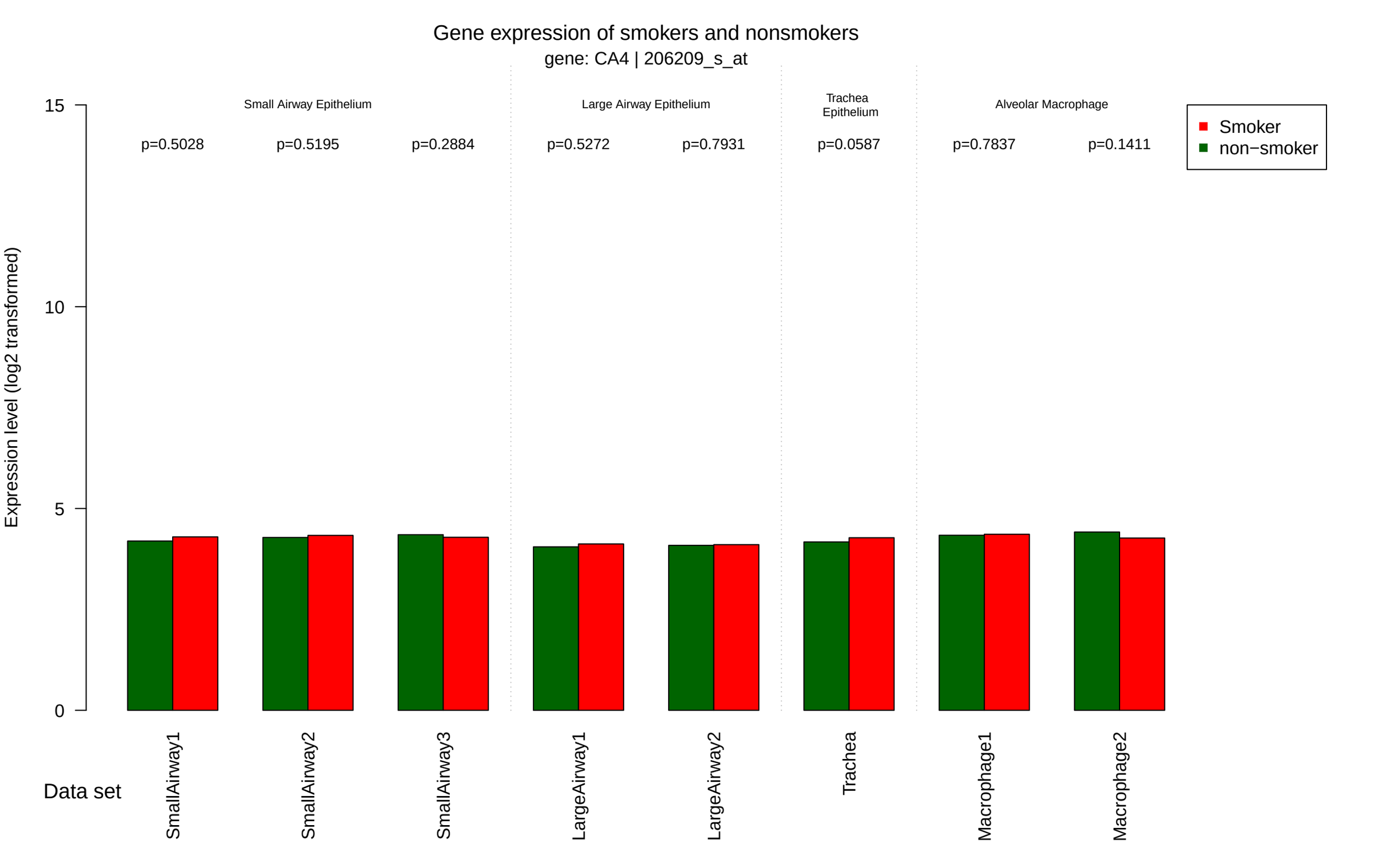
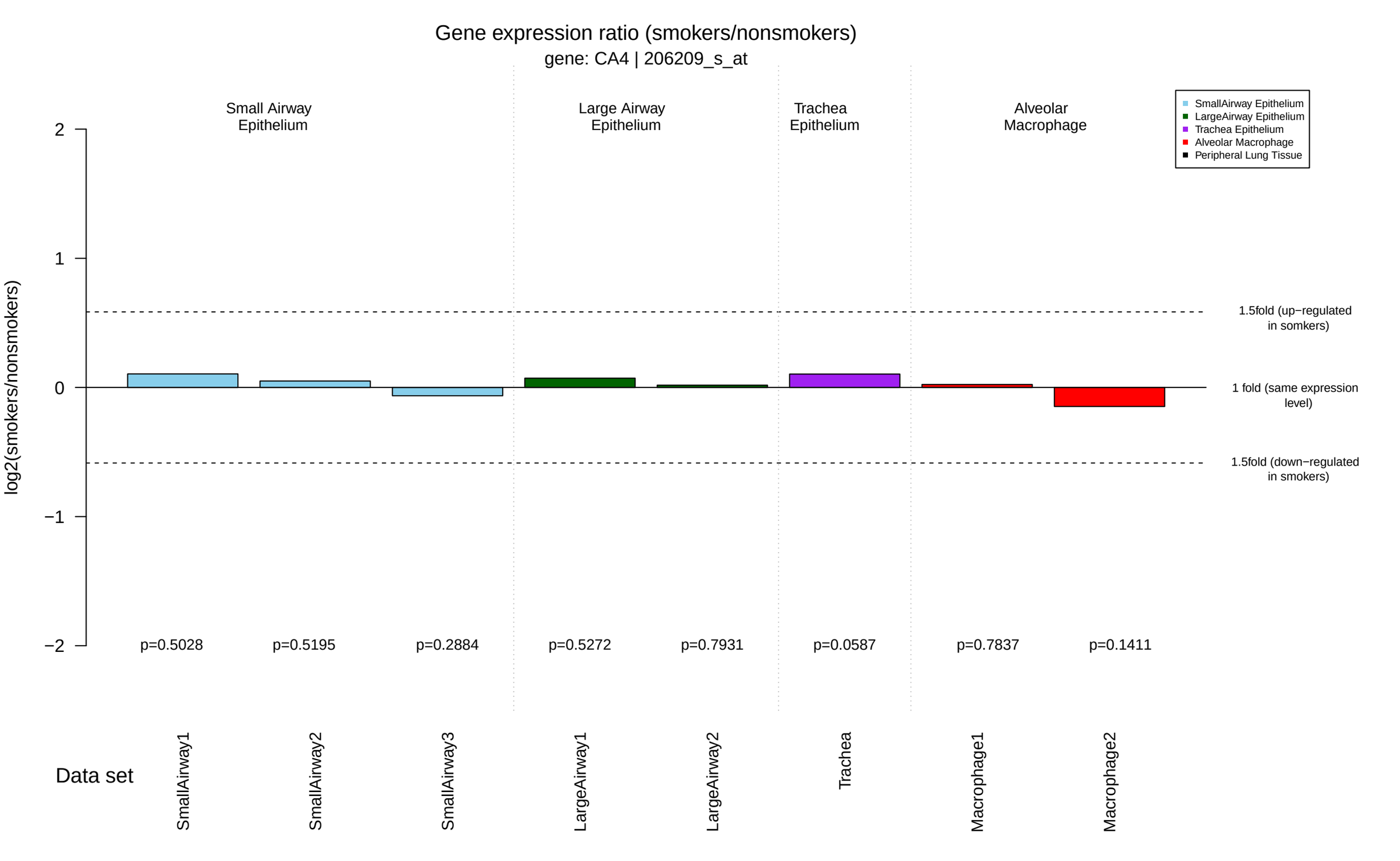
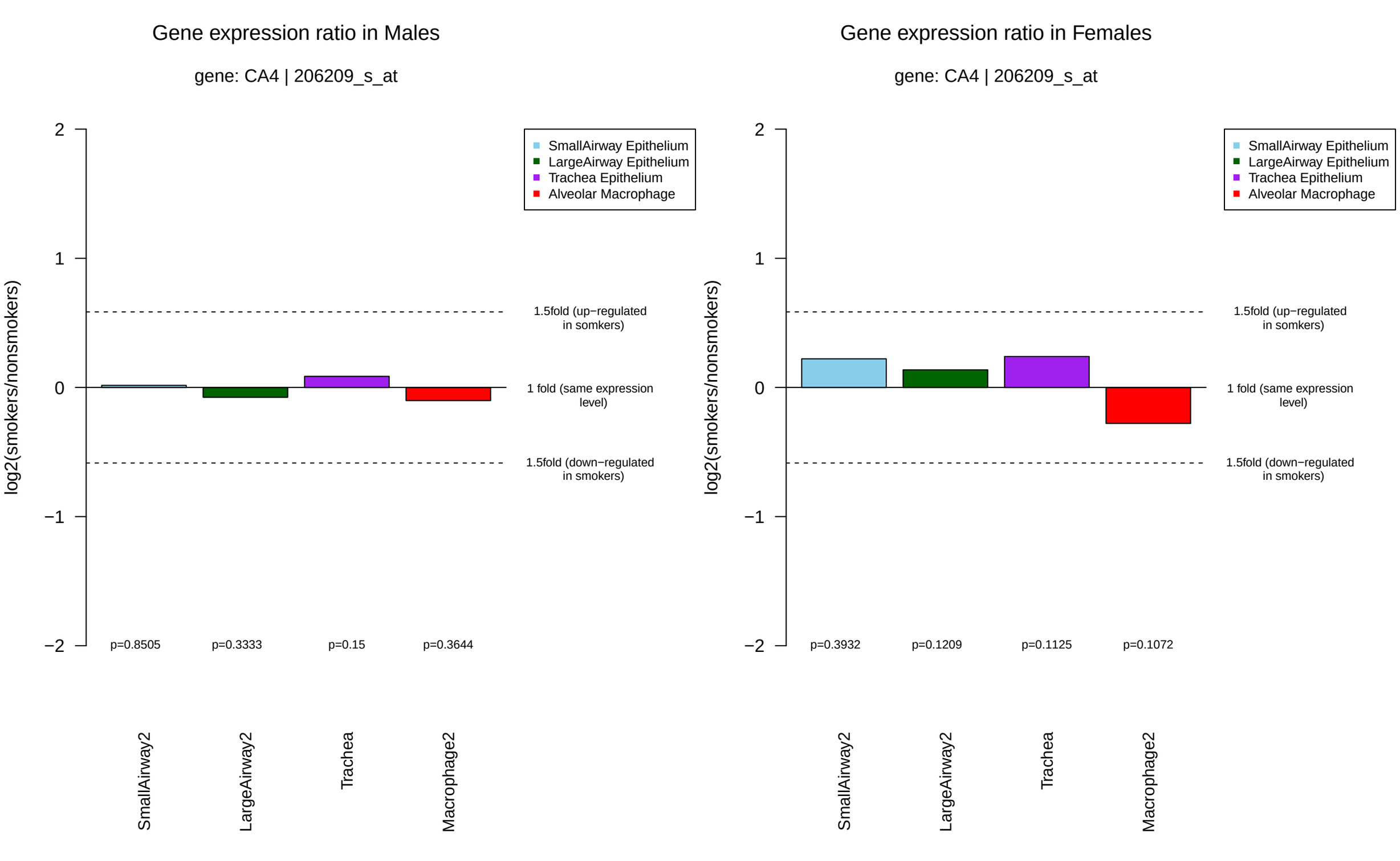
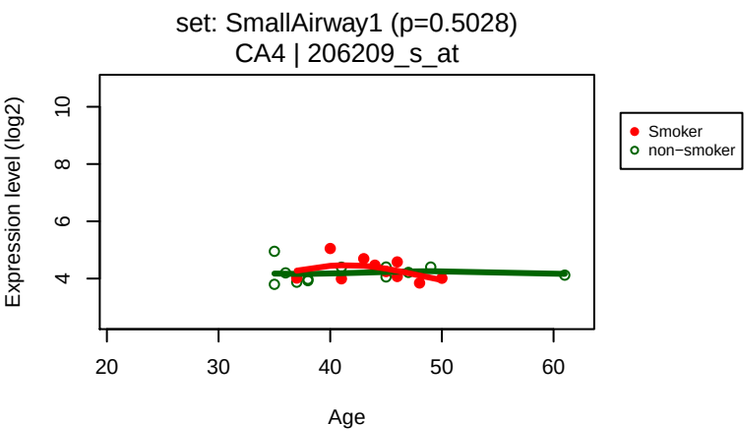
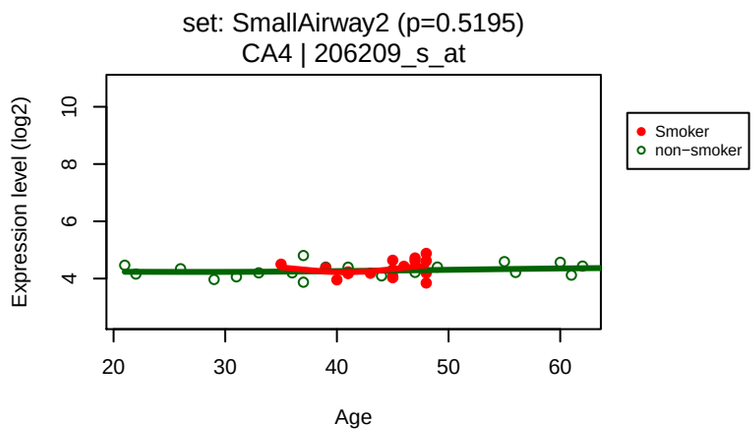
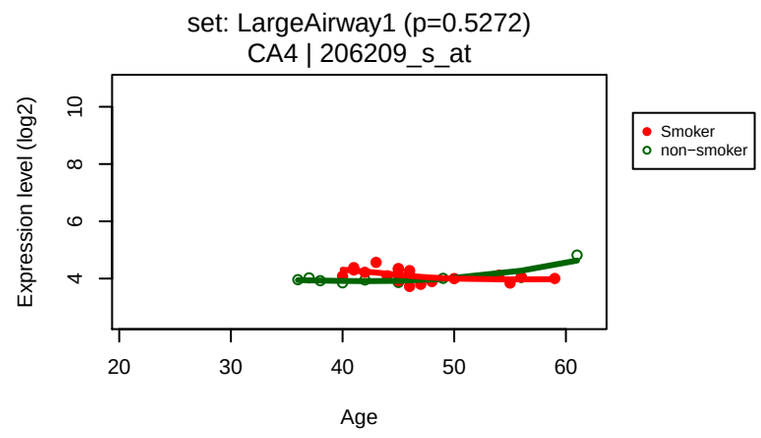

Gene expression of smokers and nonsmokers
gene: CA4 | 206209_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5028
p=0.5195
p=0.2884
p=0.5272
p=0.7931
p=0.0587
p=0.7837
p=0.1411
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

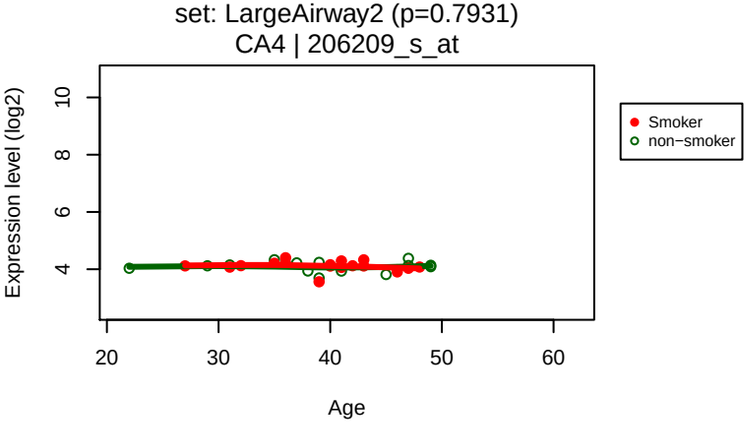

Gene expression ratio (smokers/nonsmokers)
gene: CA4 | 206209_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5028
p=0.5195
p=0.2884
p=0.5272
p=0.7931
p=0.0587
p=0.7837
p=0.1411
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

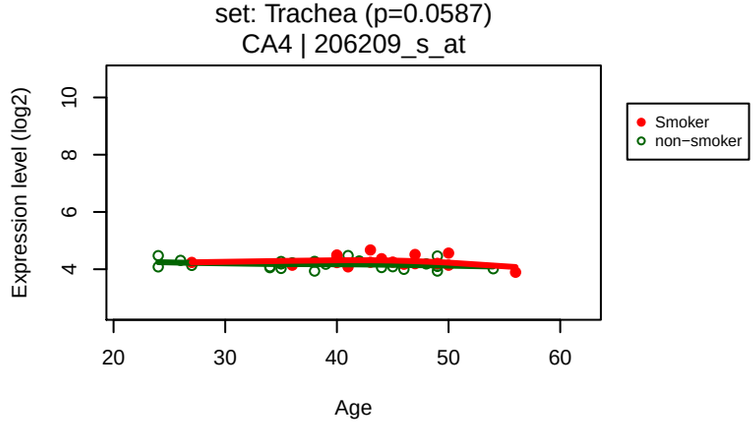

Gene expression ratio in Males
gene: CA4 | 206209_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8505
p=0.3333
p=0.15
p=0.3644
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CA4 | 206209_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3932
p=0.1209
p=0.1125
p=0.1072
SmallAirway2
LargeAirway2
Trachea
Macrophage2

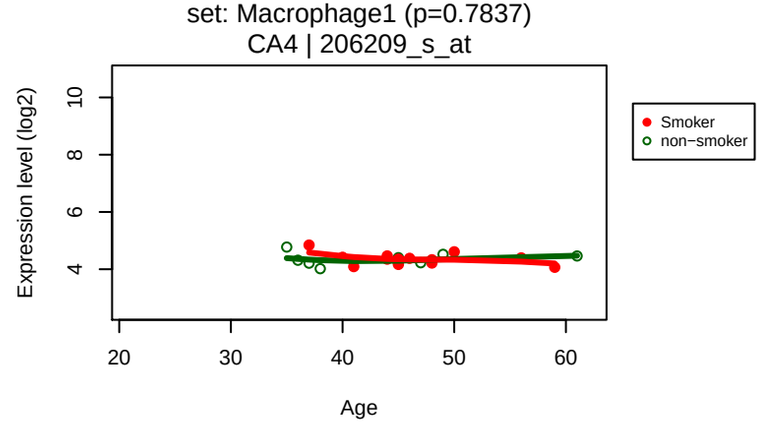

set: SmallAirway1 (p=0.5028)
CA4 | 206209_s_at
Expression level (log2)
Age
Smoker
non−smoker

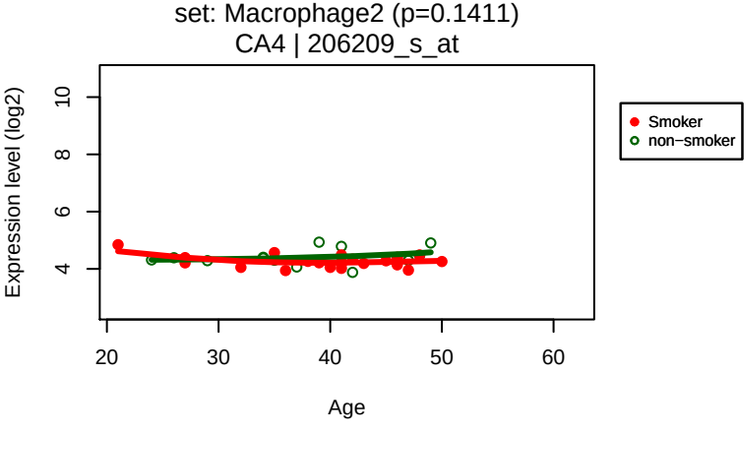

set: SmallAirway2 (p=0.5195)
CA4 | 206209_s_at
Expression level (log2)
Age
Smoker
non−smoker

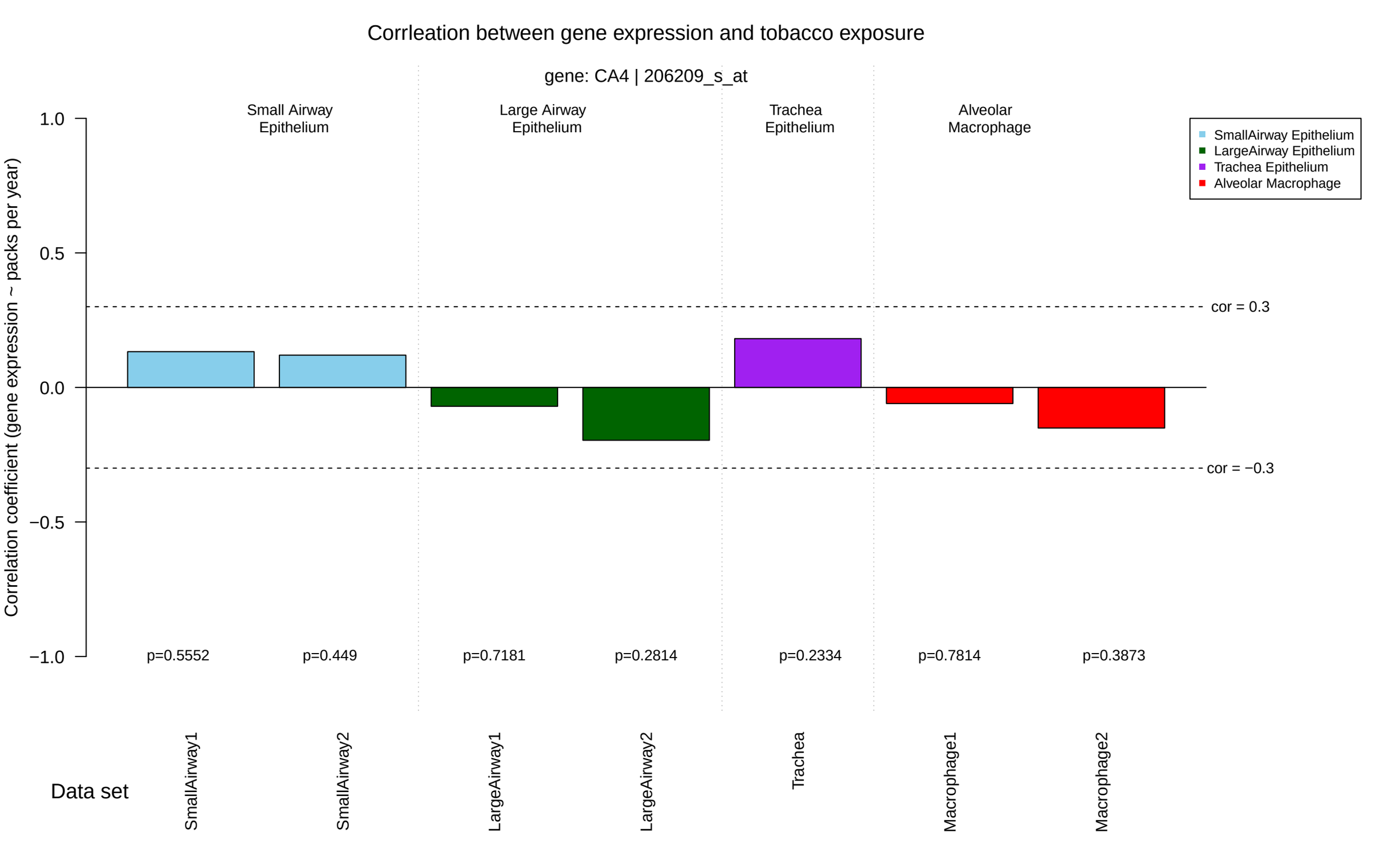

set: LargeAirway1 (p=0.5272)
CA4 | 206209_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7931)
CA4 | 206209_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0587)
CA4 | 206209_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7837)
CA4 | 206209_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1411)
CA4 | 206209_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CA4 | 206209_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5552
p=0.449
p=0.7181
p=0.2814
p=0.2334
p=0.7814
p=0.3873
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2